BIRZEIT UNIVERSITY

Faculty of Engineering and Technology
Department of Electrical and Computer Engineering

ENCS3330
Digital Integrated Circuits

HW1 – Assignment #1

Design and Simulation of CMOS Logic Gates

Prepared by:	Karim Taha
Student Number:	1211155
Instructor:	Dr. Alhareth Zyoud
Section:	1
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Abstract

Three basic CMOS logic gates are designed and simulated in this report: the two-input NAND gate (NAND2), the inverter, and the two-input NOR gate (NOR2). LTspice was used to implement each circuit at both the schematic and layout levels. In addition to measuring rise time, fall time, and propagation delay, simulations were run to confirm proper logic behavior. The impact of layout parasitics on circuit performance is demonstrated by a comparison of schematic and layout results.

Contents

1. Theory

1.1 NMOS Transistor

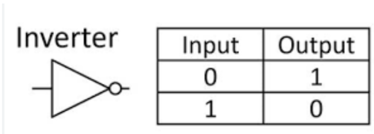
The NMOS (N-channel Metal-Oxide-Semiconductor) transistor conducts when a HIGH voltage is applied to its gate. It is used as the pull-down device in CMOS circuits, connecting the output to GND when active. NMOS transistors are fast because current is carried by electrons, which have high mobility.

1.2 PMOS Transistor

The PMOS (P-channel Metal-Oxide-Semiconductor) transistor conducts when a LOW voltage is applied to its gate. It is used as the pull-up device in CMOS circuits, connecting the output to V_{DD} when active. PMOS transistors are slower than NMOS because current is carried by holes, which have approximately 2–3 times lower mobility than electrons.

1.3 CMOS Inverter

The CMOS inverter consists of one PMOS (pull-up) and one NMOS (pull-down) transistor. When input A is HIGH, the NMOS conducts and pulls the output LOW. When input A is LOW, the PMOS conducts and pulls the output HIGH. No static current flows in either steady state, so static power dissipation is zero.

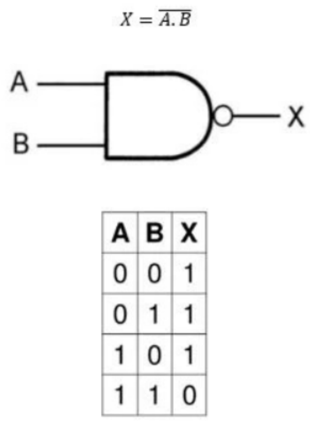


Input	Output
0	1
1	0

Figure 1: Inverter gate symbol and truth table.

1.4 2-Input NAND Gate

The NAND2 gate implements $Y = \overline{A \cdot B}$. Its pull-down network consists of two NMOS transistors in series, and its pull-up network consists of two PMOS transistors in parallel. The output is LOW only when both inputs A and B are simultaneously HIGH.

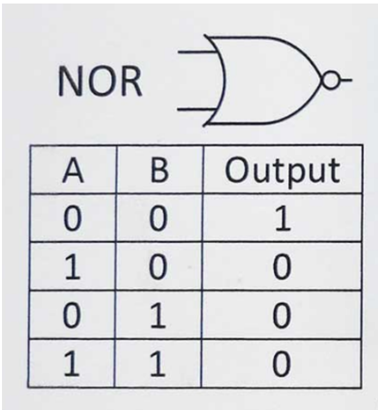


A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

Figure 2: NAND2 gate symbol and truth table.

1.5 2-Input NOR Gate

The NOR2 gate implements $Y = \overline{A+B}$. Its pull-up network consists of two PMOS transistors in series, and its pull-down network consists of two NMOS transistors in parallel. The output is HIGH only when both inputs A and B are simultaneously LOW.



A	B	Output
0	0	1
1	0	0
0	1	0
1	1	0

Figure 3: NOR2 gate symbol and truth table.

1.6 Transistor Sizing

PMOS transistors must be wider than NMOS transistors to compensate for lower hole mobility and to achieve balanced rise and fall times. For gates with series-stacked transistors, the stacked devices must be further widened to restore the drive strength lost due to the series resistance of the stack.

The general sizing rules applied in this work are as follows. For the Inverter, the PMOS width is set to $2W_n$. For NAND2, the series NMOS devices are each sized at $2W_n$ while the parallel PMOS devices remain at W_n. For NOR2, the series PMOS devices are each sized at $4W_n$ while the parallel NMOS devices remain at W_n.

1.7 Propagation Delay

Propagation delay t_{pd} is the average time for the output to respond to an input change. It is defined as:

$$t_{pd} = \frac{t_{pLH} + t_{pHL}}{2} \tag{1}$$

where t_{pLH} is the low-to-high delay and t_{pHL} is the high-to-low delay, both measured at the 50 % crossing of the output waveform. Rise time t_r is measured from 10 % to 90 % of V_{DD}, and fall time t_f from 90 % to 10 % of V_{DD}. Layout simulations yield higher delay values than schematic simulations because the extracted netlist includes parasitic capacitances from metal interconnects and diffusion regions.

2. CMOS Inverter

2.1 Schematic Design and Simulation

The PMOS source is connected to V_{DD} and the NMOS source to GND. Both gates are driven by input A, and both drains connect to output Y. The PMOS width is set to $2W_n$ for balanced rise and fall transitions.

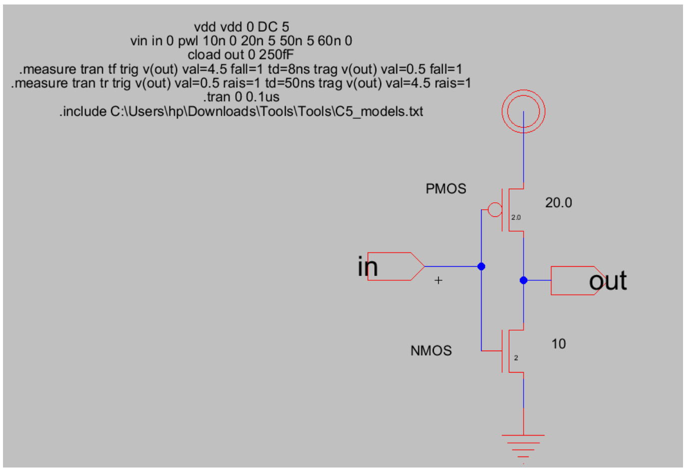


Figure 4: CMOS inverter schematic.

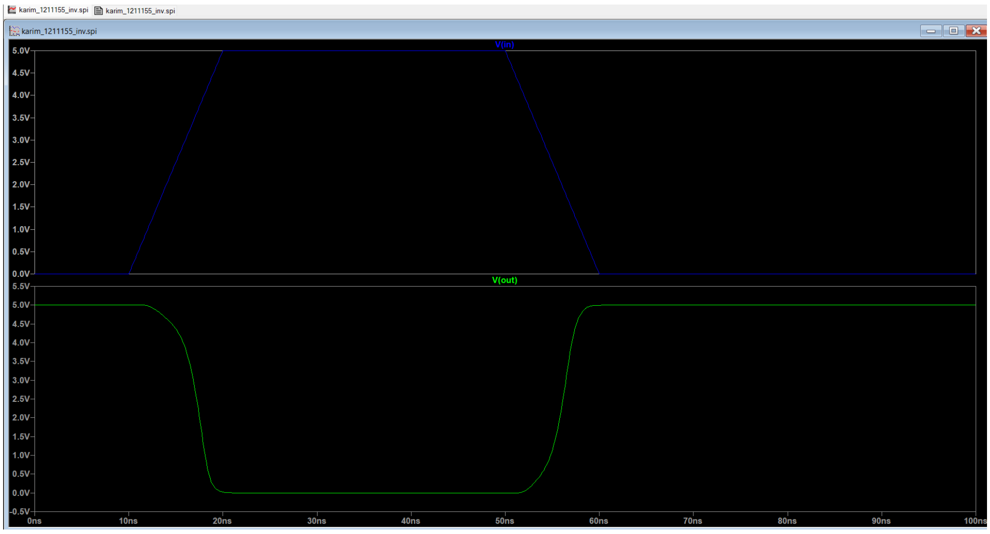


Figure 5: Inverter schematic simulation waveforms.

2.2 Layout Design and Simulation

The PMOS transistor is placed in the N-well at the top of the cell abutting the V_{DD} rail. The NMOS transistor is placed in the P-substrate at the bottom abutting the GND rail. Metal-1 connects both drain contacts to form output node Y.

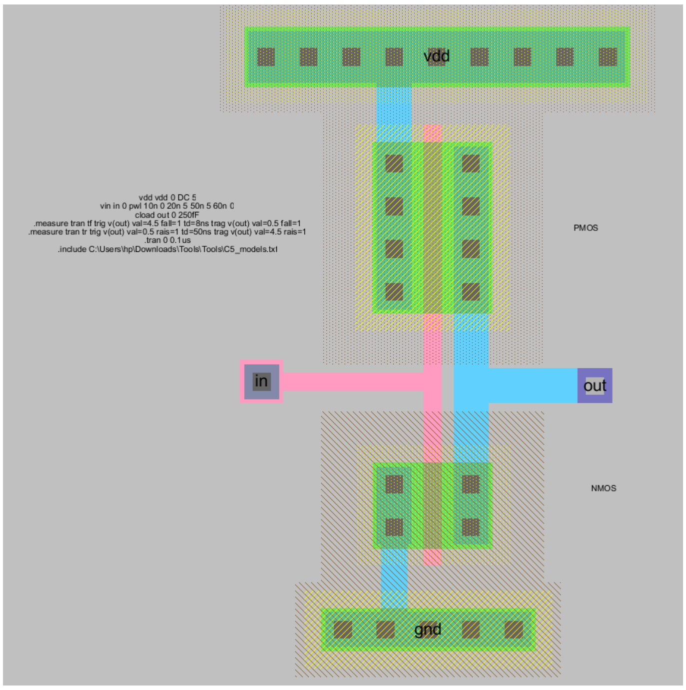


Figure 6: CMOS inverter layout.

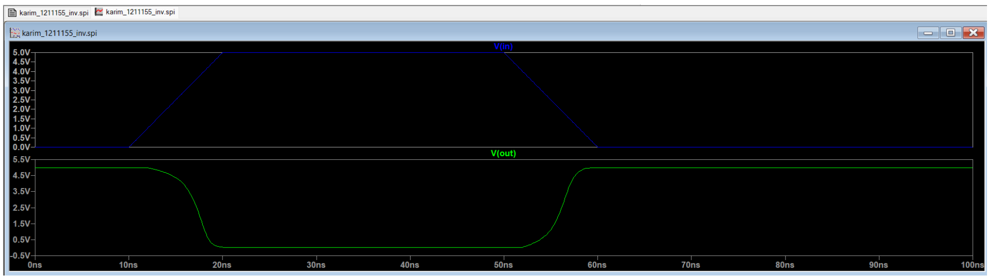


Figure 7: Inverter layout simulation waveforms.

3. 2-Input NAND Gate (NAND2)

3.1 Schematic Design and Simulation

NMOS transistors M_1 and M_2 are stacked in series (each $2W_n$) between the output and GND. PMOS transistors M_3 and M_4 are connected in parallel (W_n each) between V_{DD} and the output.

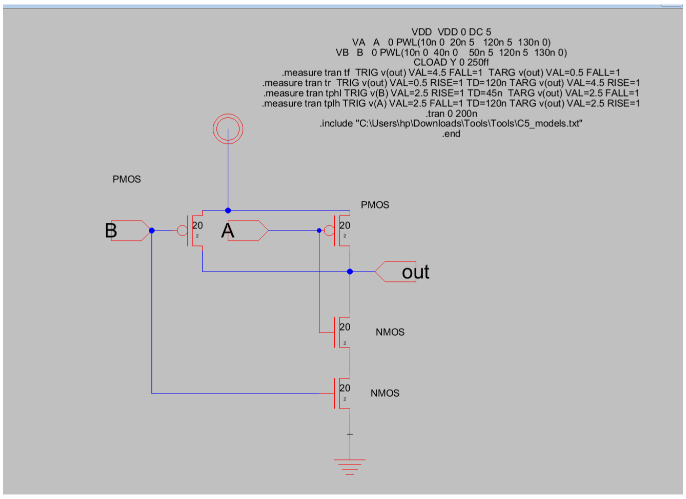


Figure 8: NAND2 schematic.

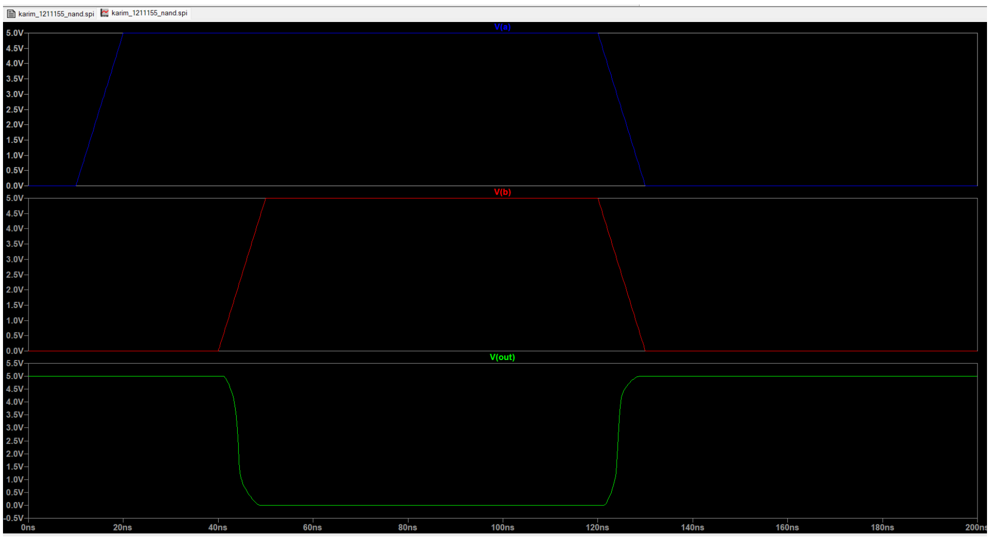


Figure 9: NAND2 schematic simulation waveforms.

3.2 Layout Design and Simulation

The two series NMOS devices are placed adjacent in the active region so that the shared internal drain/source node requires only a short Metal-1 connection. The parallel PMOS devices share the V_{DD} rail at the top of the cell.

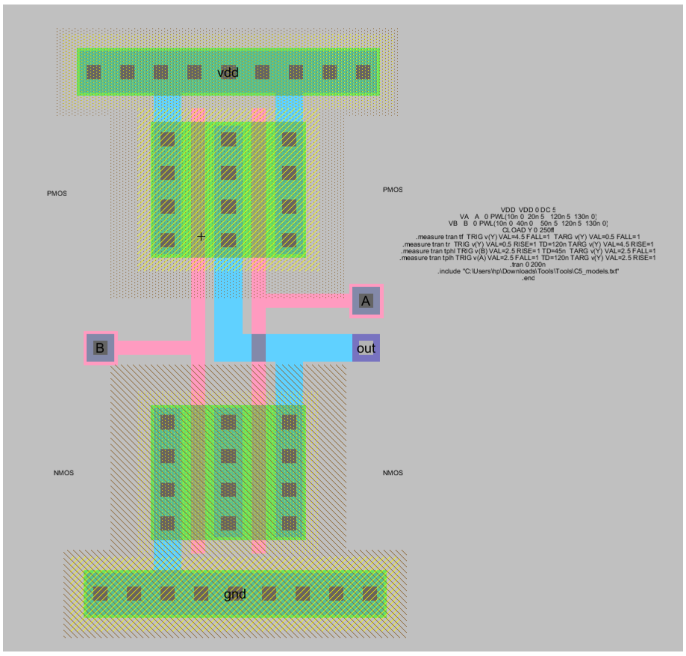


Figure 10: NAND2 layout.

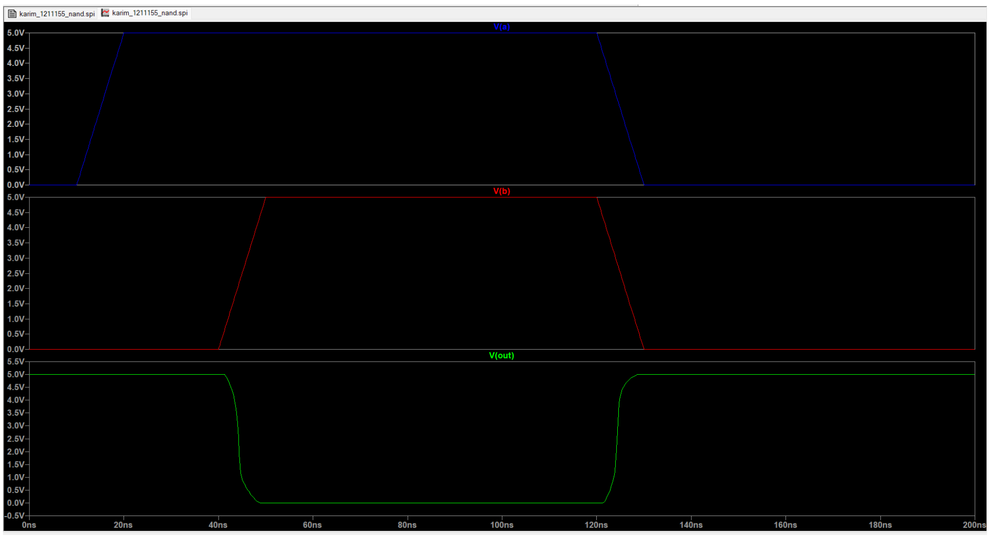


Figure 11: NAND2 layout simulation waveforms.

4. 2-Input NOR Gate (NOR2)

4.1 Schematic Design and Simulation

PMOS transistors M_3 and M_4 are stacked in series ($4W_n$ each) between V_{DD} and the output. NMOS transistors M_1 and M_2 are in parallel (W_n each) between the output and GND. The large PMOS width is necessary to compensate for the combined penalty of low hole mobility and series stacking resistance.

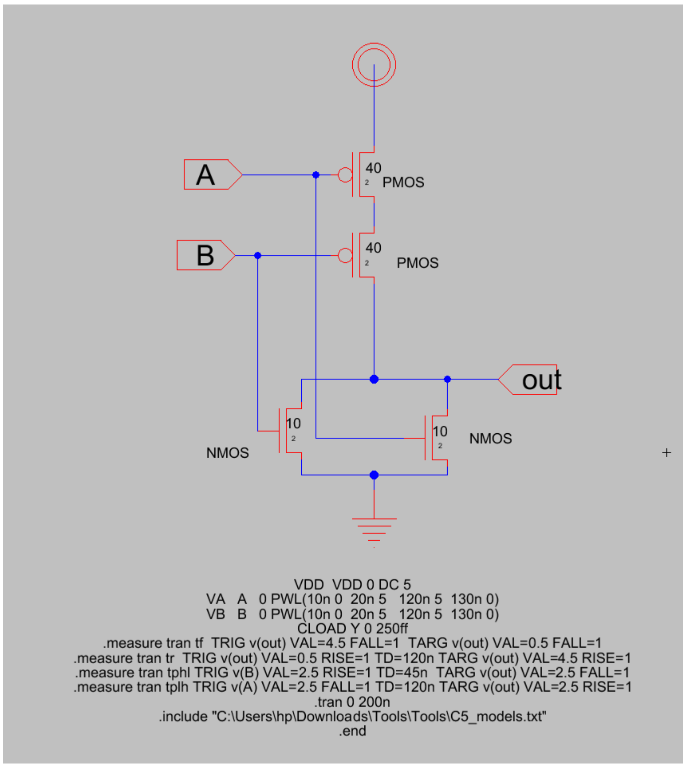


Figure 12: NOR2 schematic.

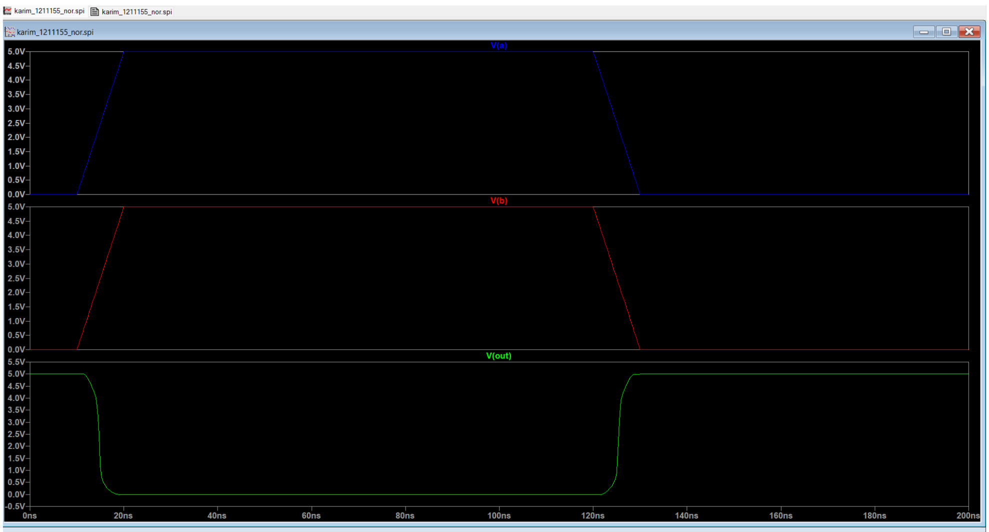


Figure 13: NOR2 schematic simulation waveforms.

4.2 Layout Design and Simulation

The series PMOS pair is placed in the N-well. The internal node between the two series PMOS transistors is kept as short as possible to minimise its parasitic capacitance, since any additional capacitance at that node directly degrades rise time.

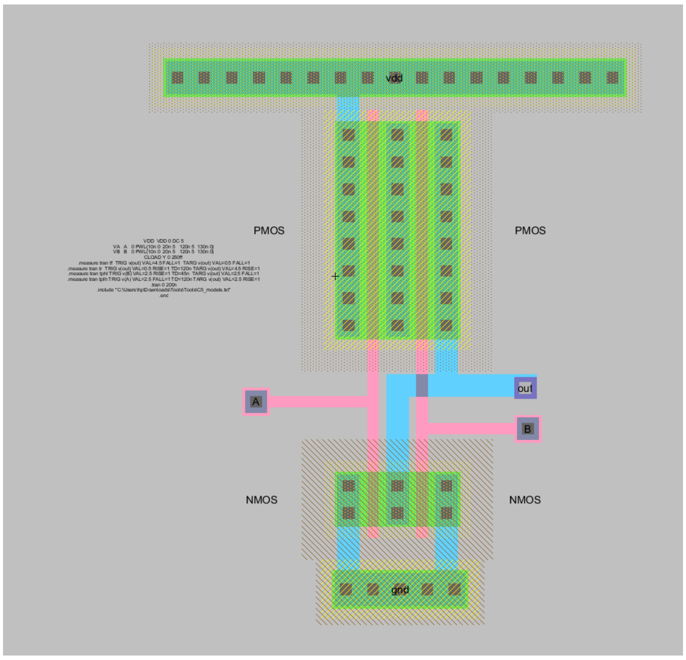


Figure 14: NOR2 layout.

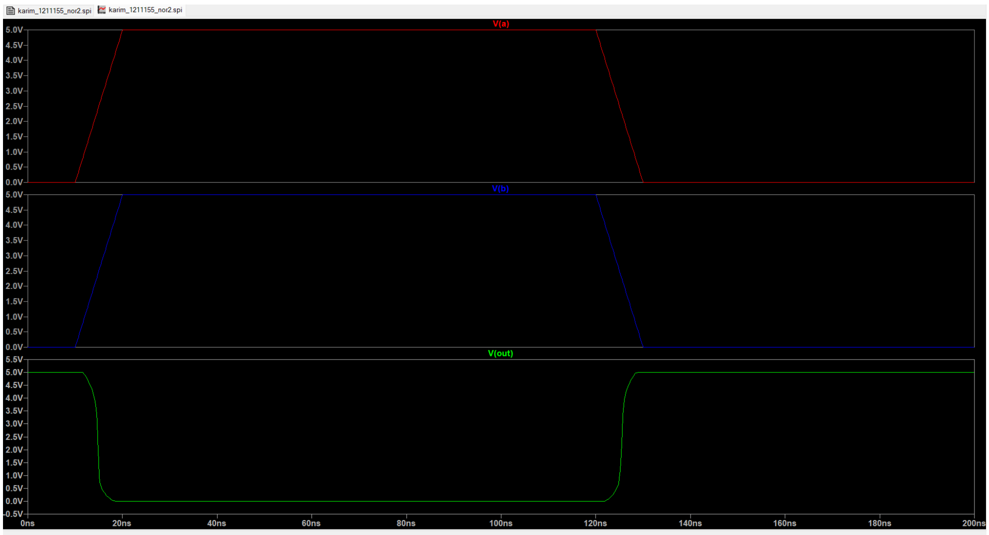
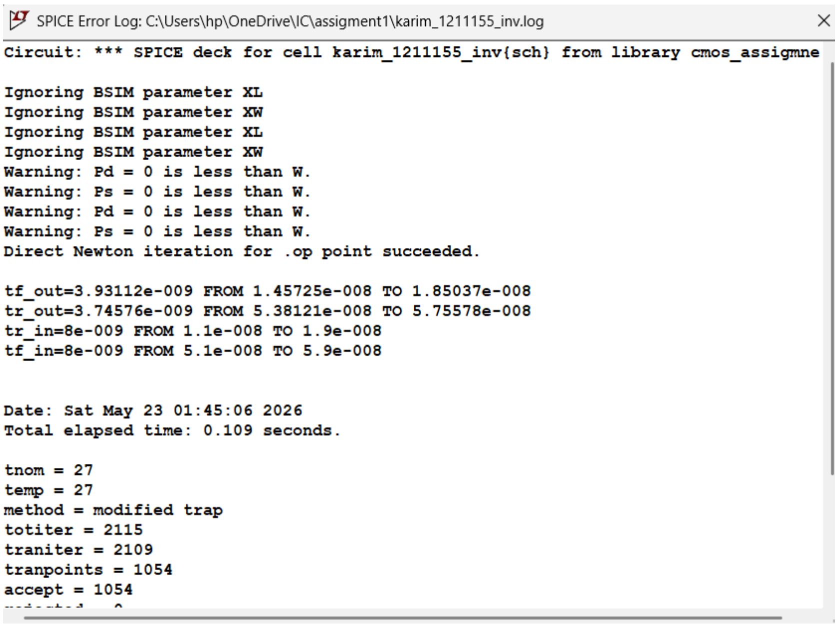


Figure 15: NOR2 layout simulation waveforms.

5. Timing Results and Comparison

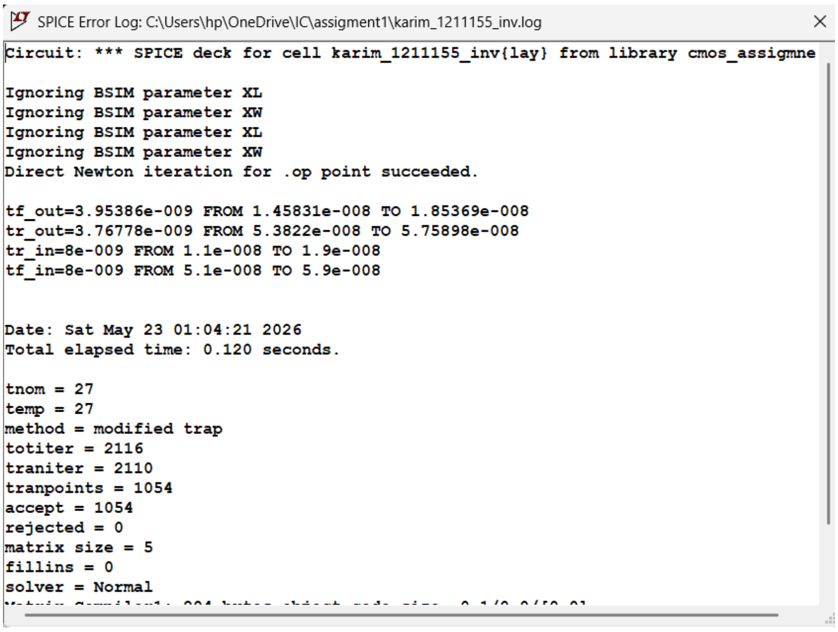
Rise time (t_r) and fall time (t_f) are measured on the output signal between the 10 % and 90 % voltage crossings. Propagation delays t_{pHL} and t_{pLH} are measured at the 50 % crossing. All measurements are taken from both the schematic and post-layout extracted simulations to quantify the effect of parasitic capacitances introduced by physical interconnects.

5.1 Inverter – Rise and Fall Times



```
SPICE Error Log: C:\Users\hp\OneDrive\IC\assigment1\karim_1211155_inv.log
Circuit: *** SPICE deck for cell karim_1211155_inv{sch} from library cmos_assigmne

Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Direct Newton iteration for .op point succeeded.

tf_out=3.93112e-009 FROM 1.45725e-008 TO 1.85037e-008
tr_out=3.74576e-009 FROM 5.38121e-008 TO 5.75578e-008
tr_in=8e-009 FROM 1.1e-008 TO 1.9e-008
tf_in=8e-009 FROM 5.1e-008 TO 5.9e-008


Date: Sat May 23 01:45:06 2026
Total elapsed time: 0.109 seconds.

tnom = 27
temp = 27
method = modified trap
totiter = 2115
traniter = 2109
tranpoints = 1054
accept = 1054
```

Figure 16: Inverter schematic – rise and fall time measurement.



```
SPICE Error Log: C:\Users\hp\OneDrive\IC\assigment1\karim_1211155_inv.log
Circuit: *** SPICE deck for cell karim_1211155_inv{lay} from library cmos_assigmne

Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Direct Newton iteration for .op point succeeded.

tf_out=3.95386e-009 FROM 1.45831e-008 TO 1.85369e-008
tr_out=3.76778e-009 FROM 5.3822e-008 TO 5.75898e-008
tr_in=8e-009 FROM 1.1e-008 TO 1.9e-008
tf_in=8e-009 FROM 5.1e-008 TO 5.9e-008


Date: Sat May 23 01:04:21 2026
Total elapsed time: 0.120 seconds.

tnom = 27
temp = 27
method = modified trap
totiter = 2116
traniter = 2110
tranpoints = 1054
accept = 1054
rejected = 0
matrix size = 5
fillins = 0
solver = Normal
```

Figure 17: Inverter layout – rise and fall time measurement.

Table 1: Inverter timing results: schematic vs. layout.

Parameter	**Schematic (ns)**	**Layout (ns)**
Rise time t_r	3.746	3.768
Fall time t_f	3.931	3.954

The schematic and layout of the inverter differ only slightly. Following layout extraction, rise time only increases by 0.6

There are very few parasitic capacitances. The rise and fall timings are almost equal (3.75 ns vs. 3.93 ns), indicating that the $2W_n$ PMOS sizing rule effectively makes up for the reduced hole mobility.

5.2 NAND2 – Rise and Fall Times

```
SPICE Error Log: C:\Users\hp\OneDrive\IC\assigment1\karim_1211155_inv.log
Circuit: *** SPICE deck for cell karim_1211155_inv{sch} from library cmos_assigmne

Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Direct Newton iteration for .op point succeeded.

tf_out=3.93112e-009 FROM 1.45725e-008 TO 1.85037e-008
tr_out=3.74576e-009 FROM 5.38121e-008 TO 5.75578e-008
tr_in=8e-009 FROM 1.1e-008 TO 1.9e-008
tf_in=8e-009 FROM 5.1e-008 TO 5.9e-008


Date: Sat May 23 01:39:27 2026
Total elapsed time: 0.115 seconds.

tnom = 27
temp = 27
method = modified trap
totiter = 2115
traniter = 2109
tranpoints = 1054
accept = 1054
```

Figure 18: NAND2 schematic – rise and fall time measurement.

```
SPICE Error Log: C:\Users\hp\OneDrive\IC\assigment1\karim_1211155_nand.log
Circuit: *** SPICE deck for cell karim_1211155_nand{lay} from library KARIM_121115

Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Direct Newton iteration for .op point succeeded.

tf_out=3.86082e-009 FROM 4.3101e-008 TO 4.69619e-008
tr_out=3.32926e-009 FROM 1.2319e-007 TO 1.26519e-007
tphl=3.34884e-010 FROM 4.5e-008 TO 4.53349e-008
tplh=2.97587e-010 FROM 1.25e-007 TO 1.25298e-007
tr_a=8e-009 FROM 1.1e-008 TO 1.9e-008
tf_a=8e-009 FROM 1.21e-007 TO 1.29e-007
tr_b=8e-009 FROM 4.1e-008 TO 4.9e-008
tf_b=8e-009 FROM 1.21e-007 TO 1.29e-007


Date: Sat May 23 01:13:45 2026
Total elapsed time: 0.087 seconds.

tnom = 27
temp = 27
method = modified trap
totiter = 2247
traniter = 2236
tranpoints = 1061
accept = 1059
```

Figure 19: NAND2 layout – rise and fall time measurement.

Table 2: NAND2 timing results: schematic vs. layout.

Parameter	**Schematic (ns)**	**Layout (ns)**
Rise time t_r	3.314	3.329
Fall time t_f	3.844	3.861
t_{pHL} (H→L)	0.305	0.335
t_{pLH} (L→H)	0.280	0.298
Propagation delay t_{pd}	0.293	0.316

Because its series stack sits on the NMOS side, where electrons convey current with great mobility, the NAND2 gate exhibits rapid propagation delays (around 0.3 ns). Following layout extraction, t_{pd} rises by 7.8% from 0.293 ns to 0.316 ns. The parasitic capacitance at the internal drain node shared by the two series NMOS transistors is the reason for the highest relative overhead of the three gates. Since the greater load capacitance at the output dominates rise and fall timings, they are only marginally impacted (<0.5,

5.3 NOR2 – Rise and Fall Times

```
SPICE Error Log: C:\Users\hp\OneDrive\IC\assigment1\karim_1211155_nor.log
Circuit: *** SPICE deck for cell karim_1211155_nor{sch} from library nor_gate

Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Direct Newton iteration for .op point succeeded.

tf_out=3.35495e-009 FROM 1.36077e-008 TO 1.69626e-008
tr_out=3.21054e-009 FROM 1.24749e-007 TO 1.2796e-007
tphl=8.94759e-010 FROM 1.5e-008 TO 1.58948e-008
tplh=1.70255e-009 FROM 1.25e-007 TO 1.26703e-007
tr_a=8e-009 FROM 1.1e-008 TO 1.9e-008
tf_a=8e-009 FROM 1.21e-007 TO 1.29e-007
tr_b=8e-009 FROM 1.1e-008 TO 1.9e-008
tf_b=8e-009 FROM 1.21e-007 TO 1.29e-007


Date: Sat May 23 01:17:27 2026
Total elapsed time: 0.124 seconds.

tnom = 27
temp = 27
method = modified trap
totiter = 2240
```

Figure 20: NOR2 schematic – rise and fall time measurement.

```
SPICE Error Log: C:\Users\hp\OneDrive\IC\assigment1\karim_1211155_nor2.log
Circuit: *** SPICE deck for cell karim_1211155_nor2{lay} from library nor_gate

Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Direct Newton iteration for .op point succeeded.

tf_out=3.37982e-009 FROM 1.36236e-008 TO 1.70034e-008
tr_out=3.24286e-009 FROM 1.24757e-007 TO 1.28e-007
tphl=9.20787e-010 FROM 1.5e-008 TO 1.59208e-008
tplh=1.72688e-009 FROM 1.25e-007 TO 1.26727e-007
tr_a=8e-009 FROM 1.1e-008 TO 1.9e-008
tf_a=8e-009 FROM 1.21e-007 TO 1.29e-007
tr_b=8e-009 FROM 1.1e-008 TO 1.9e-008
tf_b=8e-009 FROM 1.21e-007 TO 1.29e-007


Date: Sat May 23 01:19:11 2026
Total elapsed time: 0.129 seconds.

tnom = 27
temp = 27
method = modified trap
totiter = 2189
traniter = 2182
tranpoints = 1056
accept = 1054
rejected = 2
```

Figure 21: NOR2 layout – rise and fall time measurement.

Table 3: NOR2 timing results: schematic vs. layout.

Parameter	Schematic (ns)	Layout (ns)
Rise time t_r	3.211	3.243
Fall time t_f	3.355	3.380
t_{pHL} (H→L)	0.895	0.921
t_{pLH} (L→H)	1.703	1.727
Propagation delay t_{pd}	1.299	1.324

Out of the three gates, the NOR2 gate has the longest propagation delay. The primary cause is the huge $t_{pLH} \approx 1.7$ ns; pushing the output HIGH necessitates passing current through two series PMOS transistors, which are slow by nature because of low

hole mobility. This bottleneck persists even with the $4,"W_n$ scaling. Following layout extraction, t_{pd} rises by 1.9In comparison to NAND2, the overhead is lower since the parasitic addition is proportionately less significant due to the already high schematic delay.

5.4 Schematic vs. Layout – Summary

Table 4: Propagation delay and rise/fall time comparison across all three gates.

Gate	**Level**	t_r (ns)	t_f (ns)	t_{pHL} (ns)	t_{pLH} (ns)	t_{pd} (ns)
Inverter	Schematic	3.746	3.931			
	Layout	3.768	3.954			
NAND2	Schematic	3.314	3.844	0.305	0.280	0.293
	Layout	3.329	3.861	0.335	0.298	0.316
NOR2	Schematic	3.211	3.355	0.895	1.703	1.299
	Layout	3.243	3.380	0.921	1.727	1.324

Table 5: Layout overhead relative to schematic.

Gate	t_{pd} **Schematic (ns)**	t_{pd} **Layout (ns)**	**Overhead (%)**
NAND2	0.293	0.316	+7.8 %
NOR2	1.299	1.324	+1.9 %

Layout parasitics – primarily capacitances from Metal-1 interconnects, diffusion contacts, and well boundaries – increase all timing metrics relative to the ideal schematic. The NOR2 gate is the slowest due to its series PMOS pull-up stack and low hole mobility. The NAND2 shows the highest relative overhead (+7.8 %) because the parasitic capacitance at the internal NMOS node adds significantly to an already-short delay. The inverter exhibits the smallest absolute change since it contains no series-stacked devices and the shortest interconnects.

6. Conclusion

Using LTspice, the Inverter, NAND2, and NOR2 were effectively created and validated by simulation at the schematic and layout levels.

Transistor scaling was used to balance rise and fall timings: transistors in series stacks were further enlarged to restore drive strength, and PMOS widths were set larger than NMOS widths to compensate for decreased hole mobility. Simulations verified proper logic behavior for each set of inputs.

Layout parasitics increase propagation latency in all three gates, according to a comparison of schematic and layout results. For NOR2, the overhead is 1.9